

Imperfect Pitchforks and the Cusp

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Definition: Imperfection parameter

For real parameters r, h , consider

$$\dot{x} = h + rx - x^3.$$

The parameter h is the **imperfection parameter**: $h = 0$ preserves reflection symmetry, while $h \neq 0$ breaks it.

Definition: Stability diagram

A **stability diagram** partitions parameter space into regions with qualitatively different equilibrium configurations.

Definition: Generic cusp

For a C^3 scalar family $\dot{x} = f(x; \mu_1, \mu_2)$, a **generic cusp** at $(x_c; \mu_{1,c}, \mu_{2,c})$ satisfies

$$f = f_x = f_{xx} = 0, \quad f_{xxx} \neq 0, \quad \det \begin{pmatrix} f_{\mu_1} & f_{\mu_2} \\ f_{x\mu_1} & f_{x\mu_2} \end{pmatrix} \neq 0$$

at that point.

Definition: Codimension

The **codimension** of a generic bifurcation is the number of independent parameters that must be adjusted to reach it.

Observation: Codimensions of folds and cusps

A generic fold has codimension one, while a generic cusp has codimension two.

Question: How many equilibria survive a symmetry-breaking bias?

For fixed (r, h) , count the equilibria of

$$\dot{x} = h + rx - x^3.$$

Solution. Equilibria are intersections of

$$y = rx - x^3 \quad \text{and} \quad y = -h.$$

If $r \leq 0$, then

$$\frac{d}{dx}(rx - x^3) = r - 3x^2 \leq 0.$$

The cubic is strictly decreasing even when $r = 0$: its derivative vanishes only at the isolated point $x = 0$. Its values range from $+\infty$ to $-\infty$, so it crosses each horizontal line exactly once.

If $r > 0$, the cubic has turning points

$$x_{\pm} = \pm \sqrt{\frac{r}{3}}.$$

Its extremal values have magnitude

$$h_c(r) = 2 \left(\frac{r}{3} \right)^{3/2}.$$

Therefore

$$\begin{array}{l|l} |h| < h_c(r) & 3 \text{ distinct equilibria} \\ |h| = h_c(r) & 2 \text{ distinct equilibria, one double} \\ |h| > h_c(r) & 1 \text{ equilibrium.} \end{array}$$

Proposition: Fold locus of the imperfect pitchfork

The multiple-equilibrium locus of

$$\dot{x} = h + rx - x^3$$

has the parametrization

$$\boxed{r = 3s^2, \quad h = -2s^3}, \quad s \in \mathbb{R}.$$

Equivalently,

$$\boxed{27h^2 = 4r^3, \quad r \geq 0}.$$

Proof. At a multiple equilibrium $x = s$,

$$0 = f(s; r, h) = h + rs - s^3$$

and

$$0 = f_x(s; r, h) = r - 3s^2.$$

Thus

$$r = 3s^2, \quad h = s^3 - rs = -2s^3.$$

Squaring the second equation and cubing the first gives

$$h^2 = 4s^6, \quad r^3 = 27s^6,$$

so $27h^2 = 4r^3$. □

Corollary: Ordinary folds away from the cusp

For the family

$$\dot{x} = h + rx - x^3,$$

every point $(r, h) = (3s^2, -2s^3)$ with $s \neq 0$ is an ordinary saddle-node at $x = s$.

Proof. At such a point,

$$f_{xx}(s; r, h) = -6s \neq 0, \quad f_h(s; r, h) = 1 \neq 0.$$

Together with $f = f_x = 0$, these are the generic fold conditions, with h as the varying parameter and r held fixed. $\square$

Proposition: Generic cusp at the origin

The family

$$\dot{x} = h + rx - x^3$$

has a generic cusp at $(x, r, h) = (0, 0, 0)$.

Proof. At $(x, r, h) = (0, 0, 0)$,

$$f = f_x = f_{xx} = 0, \quad f_{xxx} = -6,$$

and, for parameter order (r, h) ,

$$\det \begin{pmatrix} f_r & f_h \\ f_{xr} & f_{xh} \end{pmatrix} = \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1 \neq 0.$$

Hence the defining cusp conditions hold. $\square$

Visualization: Stability diagram in parameter space

$$\mathcal{C} = \{(r, h) : 27h^2 = 4r^3, r \geq 0\}.$$

parameter region	distinct equilibria	stability
$r > 0, 27h^2 < 4r^3$	$x_1 < x_2 < x_3$	x_1, x_3 asymptotically stable; x_2 unstable
$\mathcal{C} \setminus \{(0, 0)\}$	$x_1 < x_2$	one asymptotically stable simple root; one half-stable double root
$27h^2 > 4r^3$ or $r < 0$	x_1	x_1 asymptotically stable
$(r, h) = (0, 0)$	$x^* = 0$	$\dot{x} = -x^3$: asymptotically stable

At every equilibrium,

$$\lambda = f_x(x^*; r, h) = r - 3(x^*)^2.$$

Visualization: Cusp catastrophe surface

The equilibrium surface is

$$\mathcal{S} = \{(r, h, x^*) : h = (x^*)^3 - rx^*\}.$$

A parametrization is

$$(r, s) \mapsto (r, s^3 - rs, s).$$

At fixed r , the curve $h = s^3 - rs$ has a horizontal tangent where

$$\frac{\partial h}{\partial s} = 3s^2 - r = 0.$$

These points are ordinary folds when $s \neq 0$; when $s = 0$, the condition also gives $r = 0$, yielding the cusp point. Their coordinates satisfy

$$r = 3s^2, \quad h = -2s^3, \quad 27h^2 = 4r^3.$$

operation on $\mathcal{S}$	result
$h = \text{constant}$	x^* versus r bifurcation diagram
$r = \text{constant}$	x^* versus h bifurcation diagram
projection onto (r, h)	stability diagram

Question: What happens at a specified exact fold point?

For $(r, h) = (3, 2)$, factor the vector field

$$\dot{x} = 2 + 3x - x^3$$

and classify every equilibrium.

Solution.

$$f(x) = 2 + 3x - x^3 = -(x + 1)^2(x - 2).$$

Thus $x = -1$ is a double equilibrium and $x = 2$ is a simple equilibrium. Near $x = -1$, the factor $x - 2$ is negative, so

$$f(x) > 0$$

on both sides. Hence $x = -1$ attracts from the left and repels to the right; it is half-stable. At the simple root,

$$f'(2) = 3 - 3(2)^2 = -9 < 0,$$

so $x = 2$ is asymptotically stable. The fold formula with $s = -1$ gives

$$r = 3, \quad h = 2.$$

Question: What parameters produce a prescribed double equilibrium?

For $\dot{x} = h + rx - x^3$, suppose $x = s \neq 0$ is a double equilibrium. Find (r, h) , factor the vector field, and classify the remaining root.

Solution. The equations $f(s) = f_x(s) = 0$ give

$$r = 3s^2, \quad h = -2s^3.$$

With these parameters,

$$f(x) = -(x - s)^2(x + 2s).$$

The remaining equilibrium is $x = -2s$, and

$$f'(-2s) = 3s^2 - 3(-2s)^2 = -9s^2 < 0.$$

Thus $x = -2s$ is asymptotically stable. If $s > 0$, then $h < 0$ and the fold lies on the lower arm of the cusp; if $s < 0$, then $h > 0$ and it lies on the upper arm. At the double root $x = s$, the vector field is negative on both sides when $s > 0$ and positive on both sides when $s < 0$. Hence the double root is half-stable and Lyapunov unstable.

Observation: Potential of the imperfect pitchfork

A potential for $\dot{x} = h + rx - x^3$ is

$$V(x; r, h) = -hx - \frac{r}{2}x^2 + \frac{1}{4}x^4, \quad -V_x = h + rx - x^3.$$

Observation: Landscape inside the cusp

For $V(x; r, h) = -hx - rx^2/2 + x^4/4$, parameters $r > 0$ and $27h^2 < 4r^3$ give two local minima at the asymptotically stable equilibria and an intervening local maximum at the unstable equilibrium:

$$V_{xx}(x^*; r, h) = -f_x(x^*; r, h).$$

Definition: Quasistatic parameter sweep

A **quasistatic parameter sweep** is the ideal limit of arbitrarily slow parameter variation in which the state tracks an asymptotically stable equilibrium branch for as long as that branch remains asymptotically stable.

Definition: Hysteresis

Hysteresis is path dependence in which the observed state–parameter response does not retrace the same path when the direction of parameter variation is reversed.

Observation: Hysteresis in a scalar fold system

In the present scalar fold setting, suppose a sweep begins on an asymptotically stable branch and passes through the bistable parameter region. In the quasistatic idealization, increasing and decreasing sweeps can then track different branches and jump at different fold values.

Visualization: Hysteresis at a fixed growth parameter

For $r = 3$, the folds occur at

$$(h, x^*) = (-2, 1) \quad \text{and} \quad (h, x^*) = (2, -1).$$

quasistatic sweep	tracked asymptotically stable branch	jump
$h \nearrow$	lower branch	$h = 2 : \text{lower} \rightarrow \text{upper}$
$h \searrow$	upper branch	$h = -2 : \text{upper} \rightarrow \text{lower}$

In the quasistatic limit, the tracked asymptotically stable branch ends at the fold even though the vector field remains smooth. For a finite-rate sweep, the actual jump need not occur exactly at the static fold value.